INORGANIC CUMULATIVE EXAM

October 29, 2010

In general, first row transition metal complexes $M(II)L_x^{n+}$ are high or low spin? Why?

Are there examples of metal complexes that are both low and high spin at room temperature? Explain how this is possible or why it is not possible.

If you had to "stack the cards" in favor of preparing a low spin metal complex, how would you proceed? Why?

Describe how the spin state of a metal complex is determined experimentally.

Give the Russell Saunders term symbol for Mn(II).

A high spin d^5 metal complex has very weak absorption bands in the visible spectrum with extinction coefficients that are typically smaller than 1. Why?